University of Applied Sciences

HOCHSCHULE EMDEN·LEER

Big Data - Introduction

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Agenda

1. Introduction
2. Definition of Big Data
3. Advanced Analytics
4. Data Mining
5. Business models with Big Data

Introduction

Contact points with Big Data

- What do you associate with Big Data?
- Do you know examples from your private environment at Big Data?
- Do you know any examples from your professional environment at Big Data?

Application example - in everyday life

Traffic forecasts for route optimization

- Real-time traffic analysis enables precise forecasts to be made for calculating travel times
- By storing and evaluating the movement data of countless road users, traffic jams can be predicted.

Source:

https://unsplash.com/photos/oKWHvYlKNKk

Application example - Amazon

Purchase behavior analysis of Amazon for individualized advertisement and trend analyses

- The online store Amazon collects and evaluates customer data in order to analyze and predict their purchasing behavior
- Through search queries, purchases made and the Alexa voice control, an exact profile of the customers can be created
- Targeted advertising and recommendations are created
- Pilot project "Amazon Samples": based on customer profiles, product samples are sent unsolicited to further attract customers and increase sales
- evaluated data serve likewise for the trend analysis, on which Amazon covers the demand as fast as possible and has competition advantages

Application example - in production

Production optimization with machine and sensor data

- Recording of versatile data in the manufacturing process in the context of Industry 4.0 e.g. quality-related data, times or records of tool wear
- Predictive Maintenance can use evaluated data to predict the failure of tools or machines so that timely measures can be taken to avoid downtime

Source:

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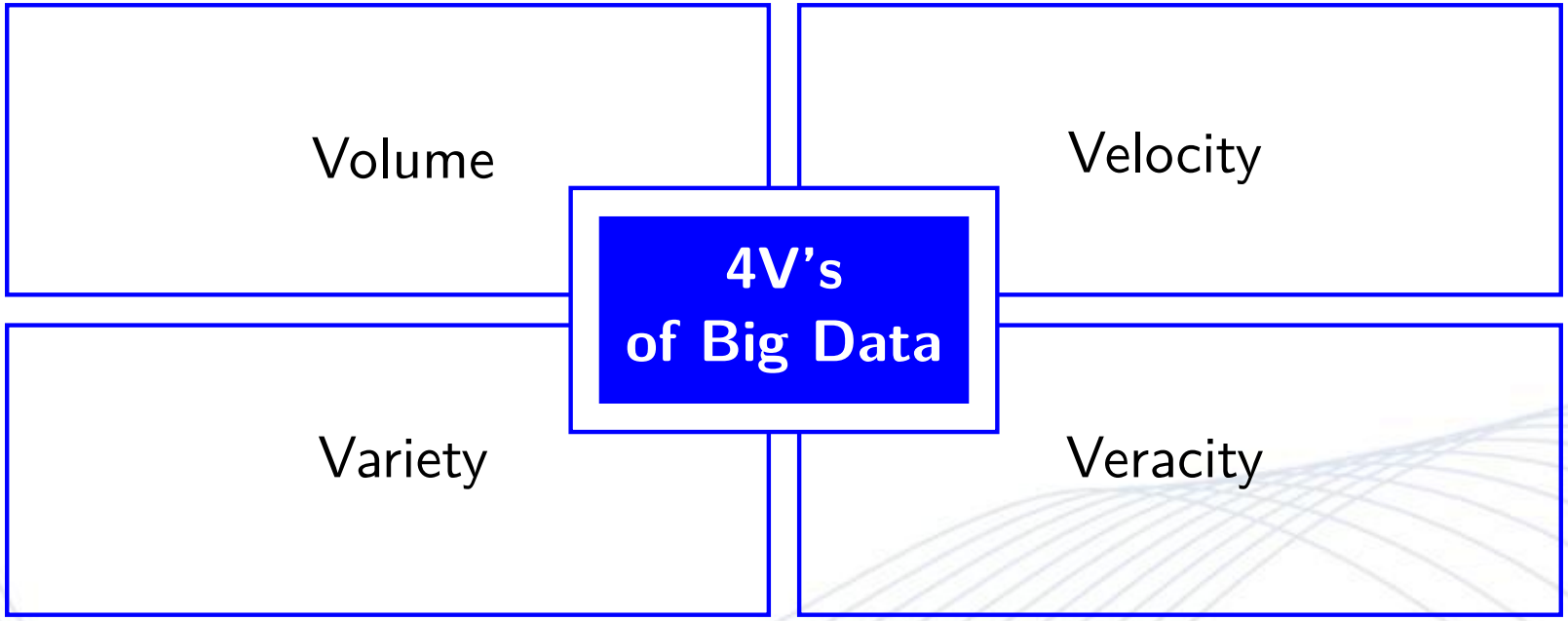
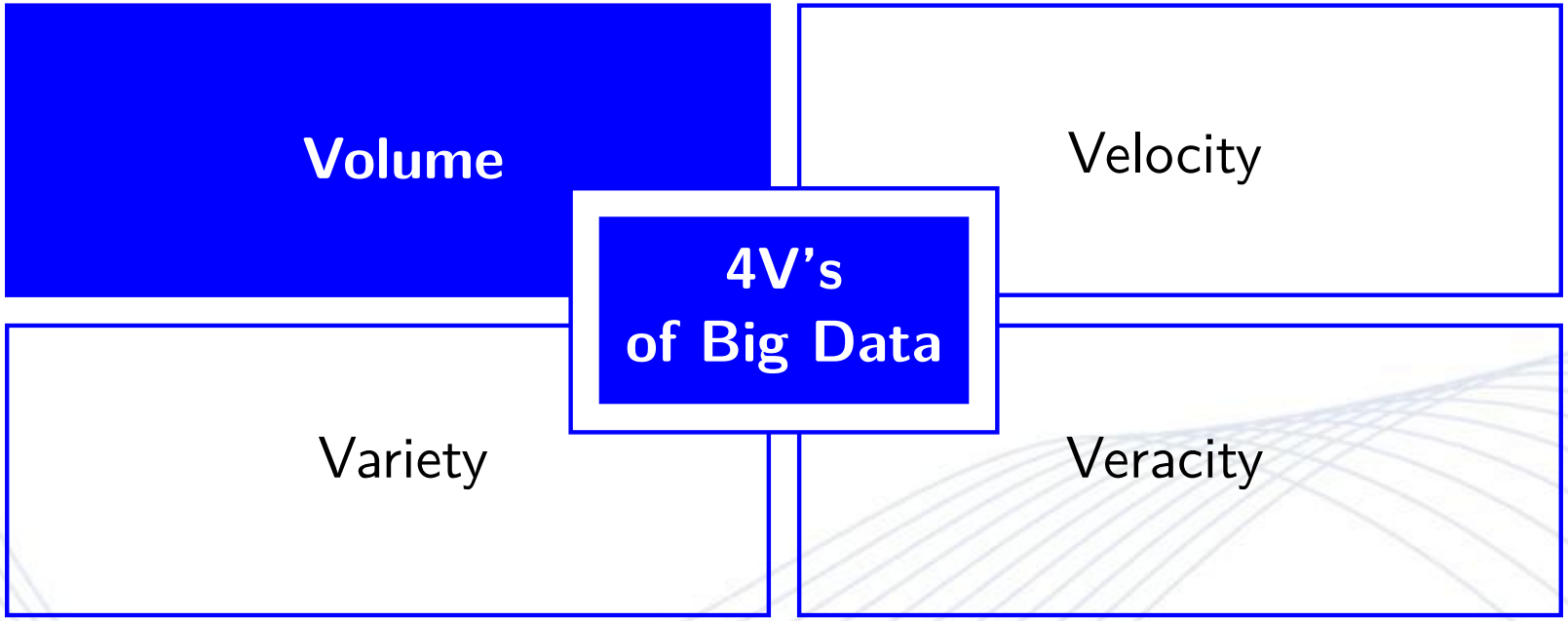
Definition of Big Data

Definition of Big Data



Big Data is described by four properties

Definition of Big Data - Volume

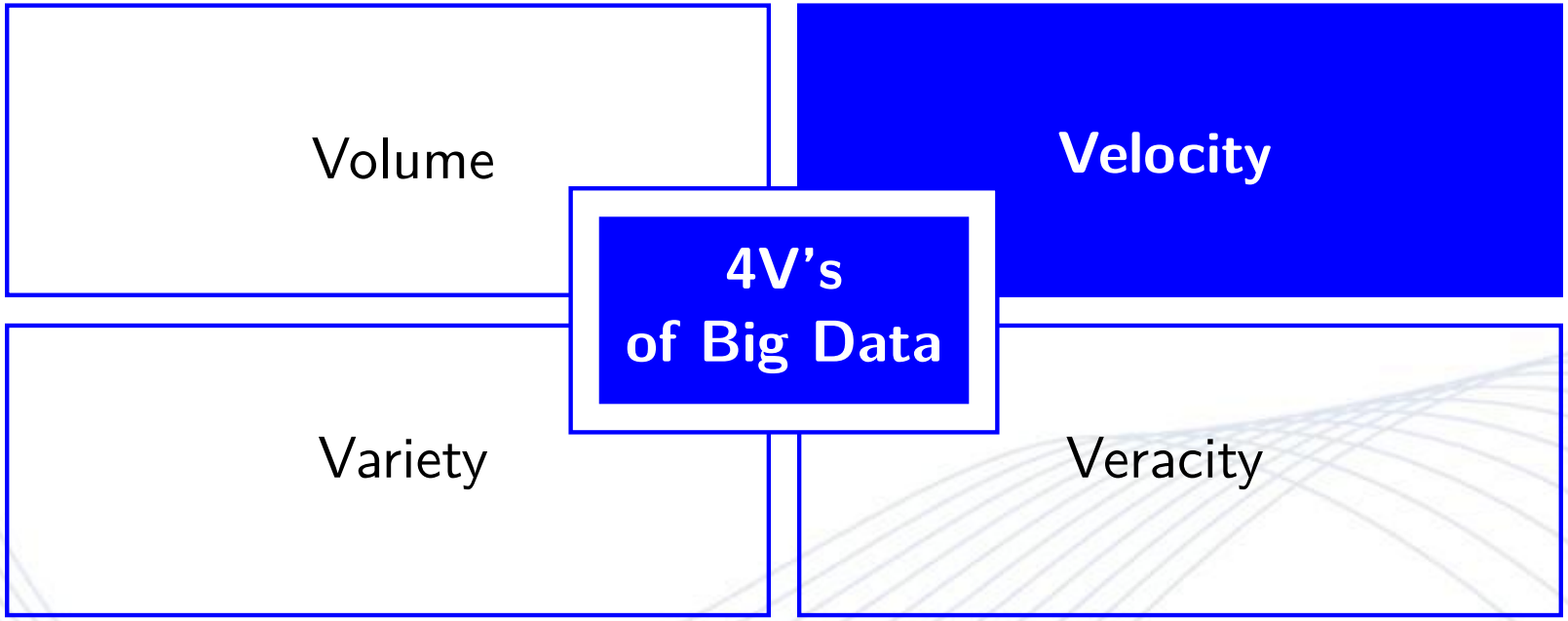


Definition of Big Data - Volume

- Extensive data volume
- Is in the tera to zettabyte range (megabyte = 10^6 bytes, gigabyte = 10^9 bytes, terabyte = 10^{12} bytes, petabyte = 10^{15} bytes, exabyte = 10^{18} bytes, zettabyte = 10^{21} bytes)
- Volume should not be considered a necessary or sufficient condition
- Managing structured data in a relational database is not considered a big data application, even if it is several petabytes in size

The basis of Big Data is the provision of more and more data for analysis

Definition of Big Data - Velocity

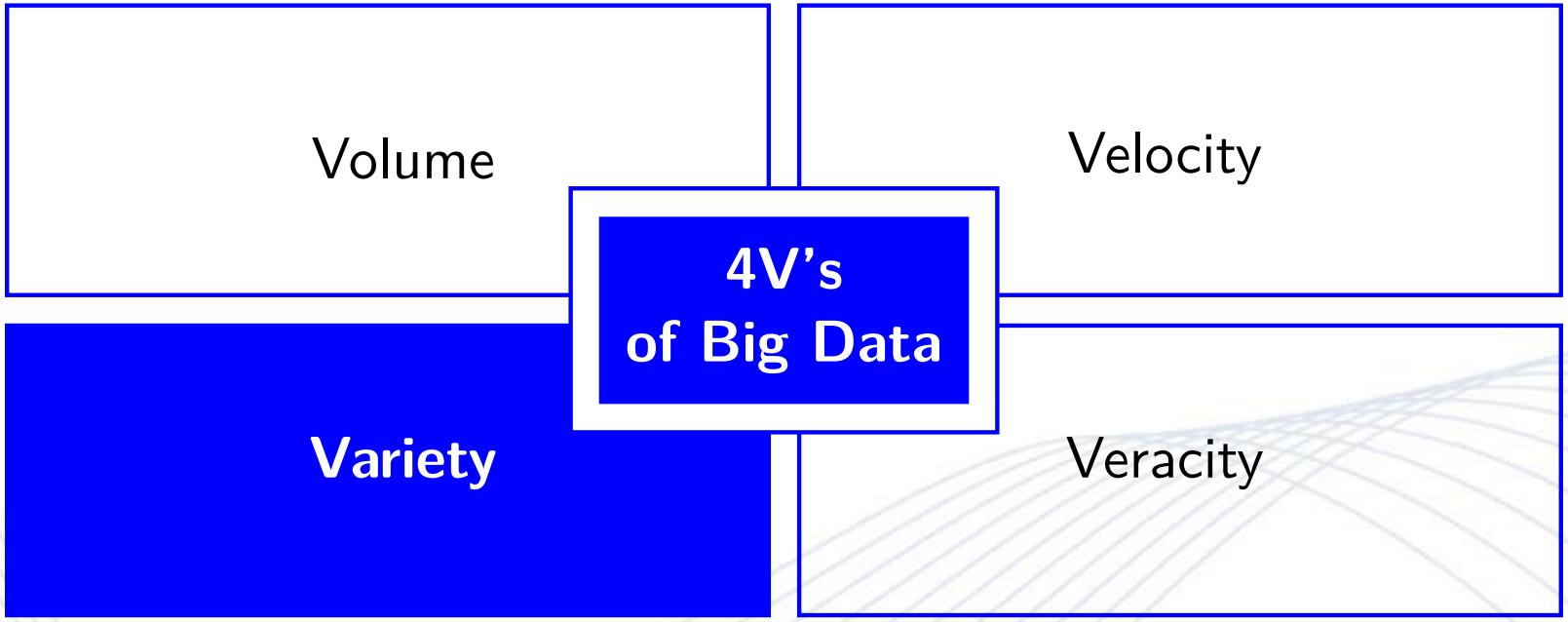


Definition of Big Data - Velocity

- Data generation at high speed
- Possibility for real-time evaluation
- Data analysis in milliseconds
- Characteristic of the short half-life of the knowledge value of data
- Quickly change facts, adding and changing data

Big Data is generated and processed at high speed

Definition of Big Data - Variety

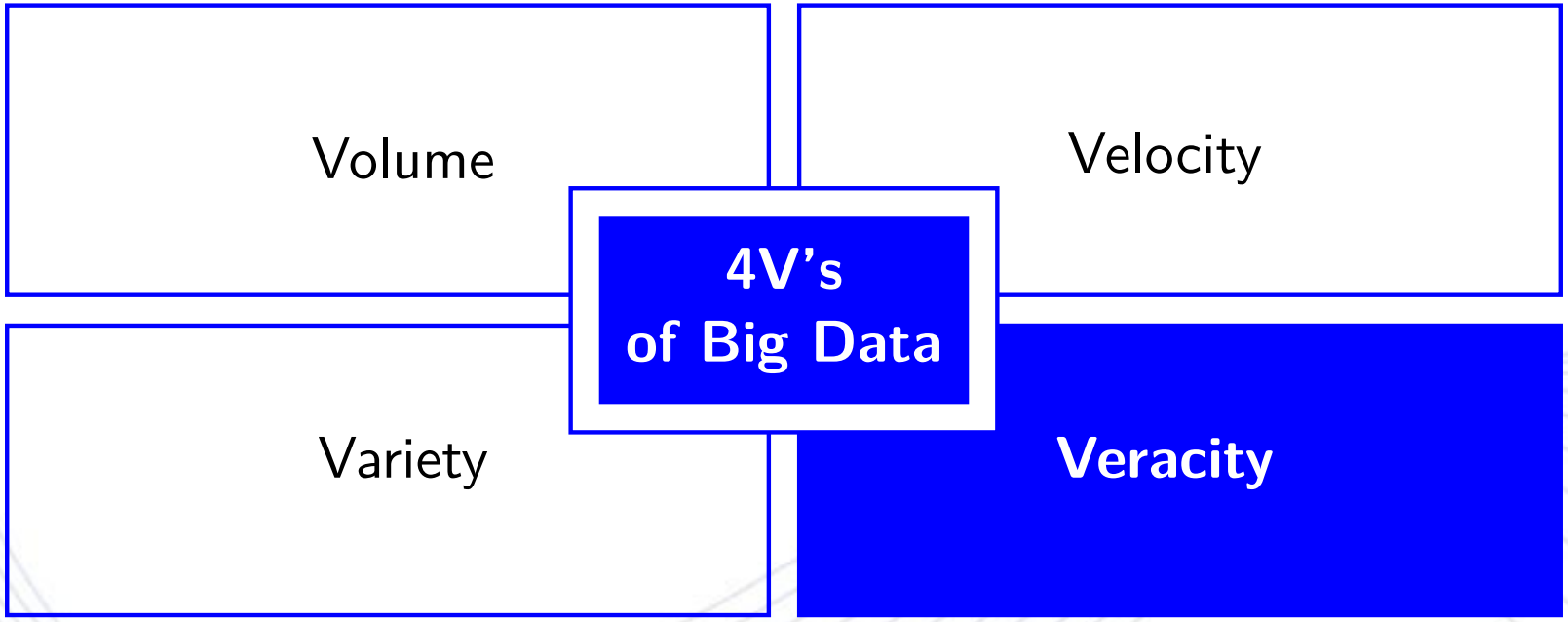


Definition of Big Data - Variety

- Heterogeneity of data source and data formats
- Polystructured data, consisting of unstructured (e.g. texts, videos, images), semi-structured (e.g. communication from and between machines) and structured (e.g. from transactional applications) data
- Predefined data models are not applicable due to the variety of data

Data occurs in great variety and is structured differently

Definition of Big Data - Veracity



Definition of Big Data - Veracity

- Data is available in varying quality
- Verification and handling of correctness, completeness and reliability of data content
- Data can be vague and inaccurate
- Specific algorithms are required to evaluate the significance of analysis results

Truth value of Big Data must be critically questioned

Challenge from Big Data

- 4 V's from Big Data: quantity, speed, variety, reliability
- Data correlations
- Machine Learning
- Forecasting and Predictive Analytics
- Intelligent data processing
- Real-time capability

To take advantage of Big Data's opportunities, challenges must be overcome

Advanced Analytics

Advanced Analytics

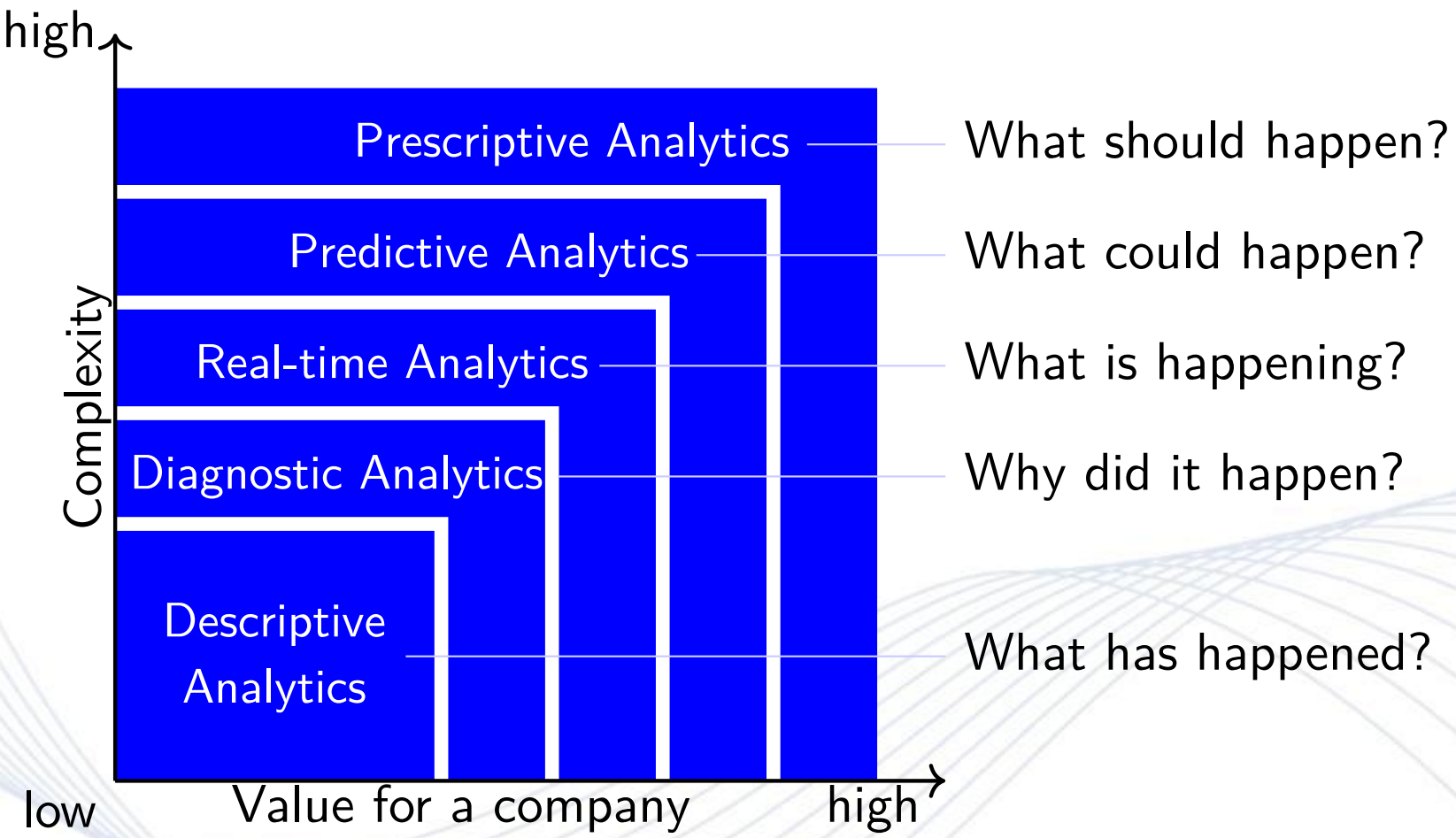
Analysis: Through a systematic examination of data, the analysis uncovers structures, anomalies or correlations and thus gains insights from data.

Analytics: The execution of analyses and the quantity of analytical methods is called analytics. Data Mining, or machine learning, as well as tools and technologies are part of the generic term.

Business Intelligence (BI): The collection, evaluation and presentation of data for operational decision support is called BI. Analytical methods are based on the strategic orientation of the company.

Advanced Analytics is suitable for processing Big Data

Variants of Analytics



Increasing complexity of analysis also results in an increase in value for companies

Data Mining

Reporting, BI analysis and data mining in comparison

	Reporting	BI analysis	Data Mining
Approach	Extract of data will be summarized accordingly.	Analysis of data to identify trends	Pattern recognition and trend prediction based on large amounts of data
Added value	**Information** Who bought the product in the last two quarters?	**Analysis** What is the average income of buyers per region of origin?	**Customer understanding** Which customers are likely to buy a similar product in the future and why?

Data Mining - Definition

Data Mining

- enables analysis of large amounts of data
- extracts information by identifying rules, patterns, trends or dependencies from Big Data
- gain insight into previously unknown and unexpected relationships between heterogeneous and large amounts of data
- Use also makes sense for smaller data volumes

Attention: Huge amounts of data due to increased sensor technology, data acquisition and storage do not yet guarantee a gain in knowledge

Data Mining - Fields of application

Computer-aided analysis methods such as artificial intelligence or statistical methods to discover "hidden" information:

- Cluster analysis
- Classification
- Regression analysis
- Forecasting methods
- Association analysis

Data Mining - Fields of application

Cluster analysis: discover similarity structures and group objects

Classification: Assignment of specific classes to individual data objects (e.g. pattern recognition to identify characteristics of employees who are likely to quit)

Regression analysis: identify relationships between dependent and independent variables

Forecasting methods: Prediction of previously unknown characteristics based on other characteristics or previously gained knowledge (e.g. predict future buying behavior from past patterns)

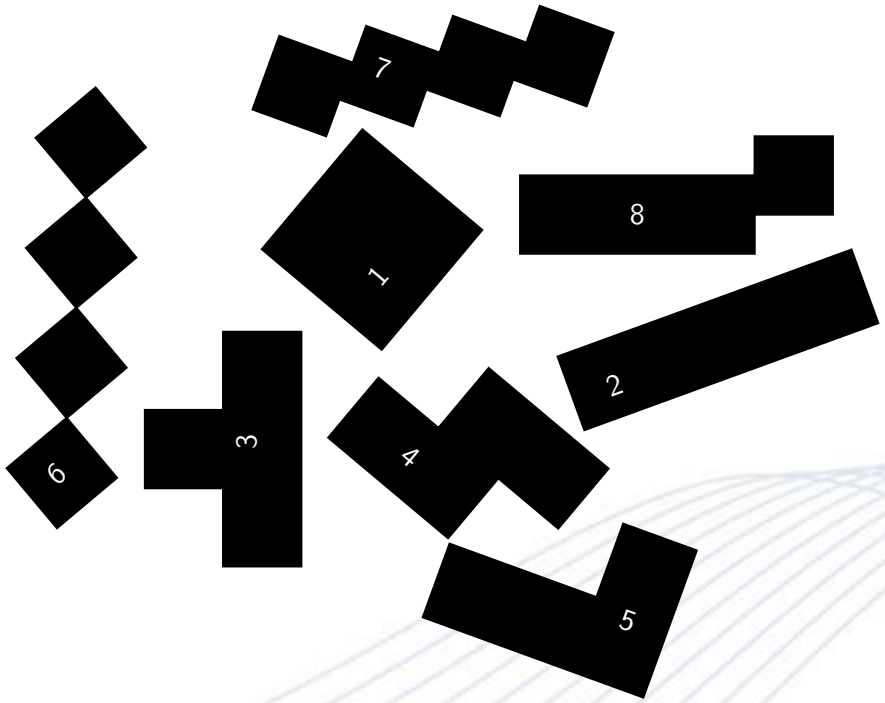
Association analysis: Find relationships between individual characteristics of an object or between different objects (e.g. connection between the purchase of beer and pizza)

Data Mining - Example of pattern recognition

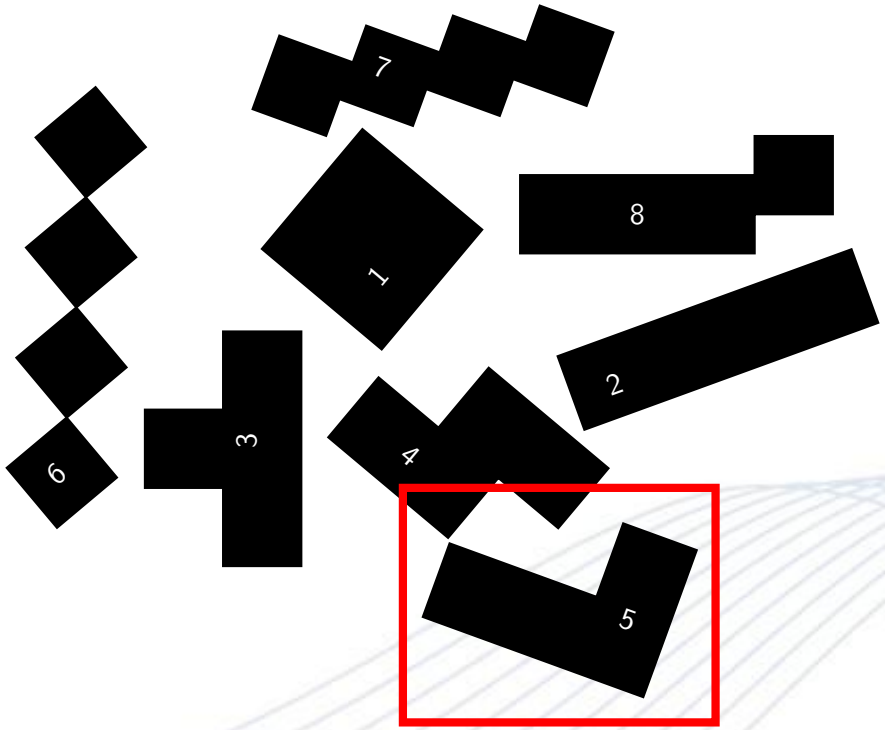
- Pattern recognition by humans: Associative ability is required
- Pattern recognition through software: Combinatorial capability predominates

$\Rightarrow$ Differences in the realization!

Pattern Recognition - Simple recognition task

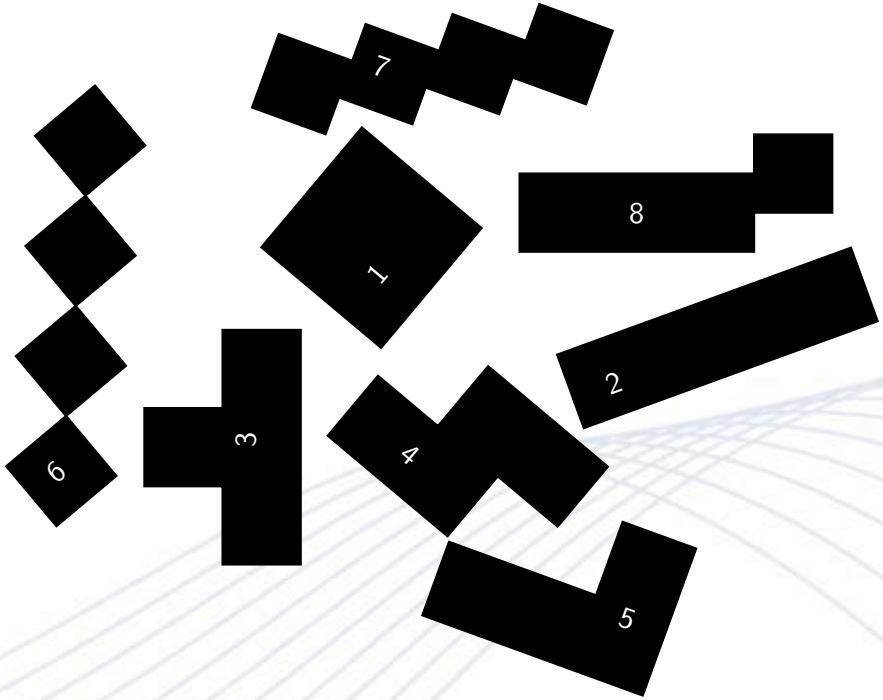


Pattern Recognition - Simple recognition task



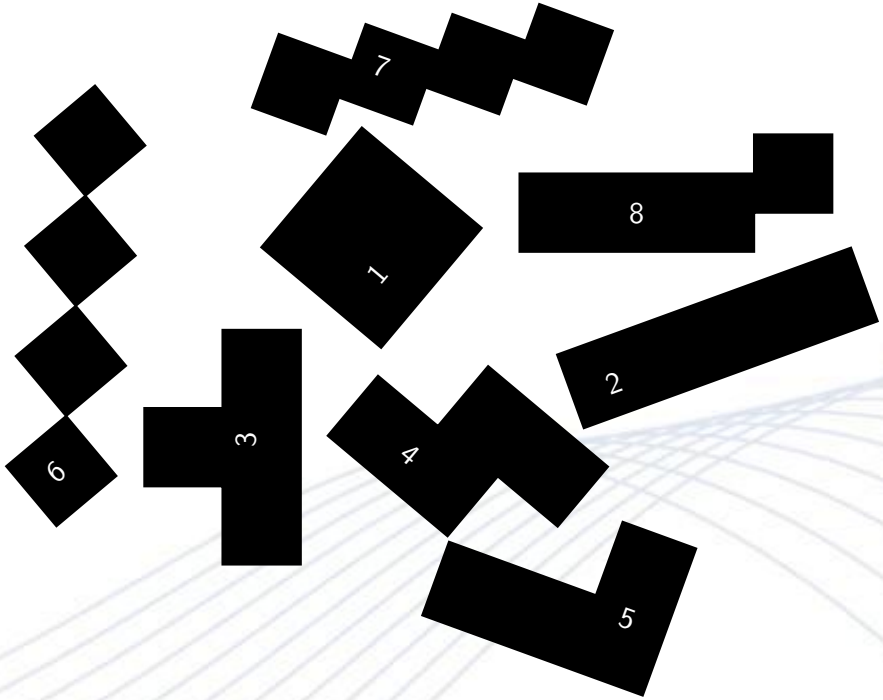
Pattern Recognition - Simple recognition task

Object	Area
1	4
2	4
3	4
4	4
5	4
6	4
7	4
8	4



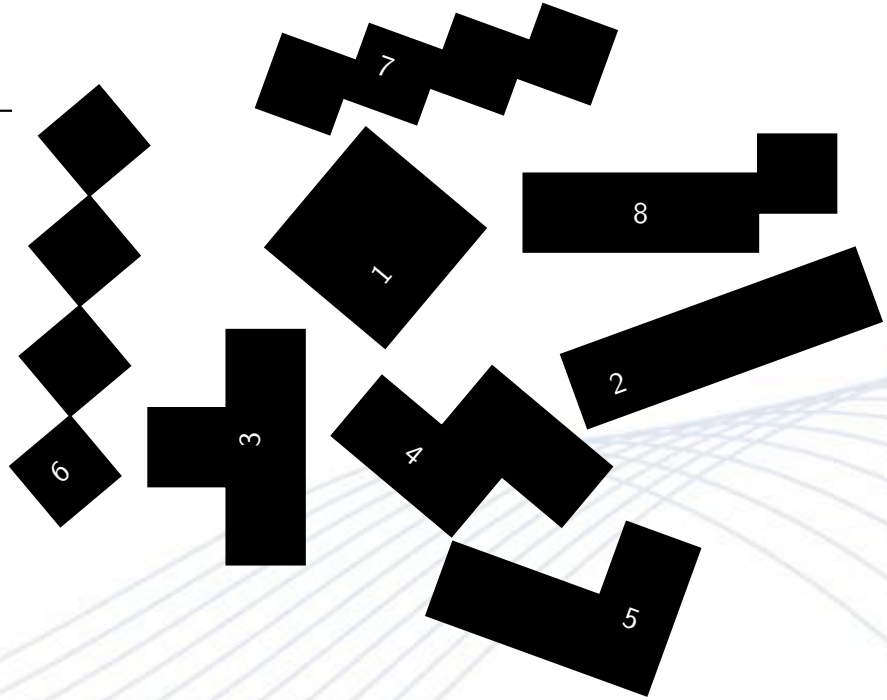
Pattern Recognition - Simple recognition task

Object	Area	Perimeter
1	4	8
2	4	10
3	4	10
4	4	10
5	4	10
6	4	16
7	4	13
8	4	11



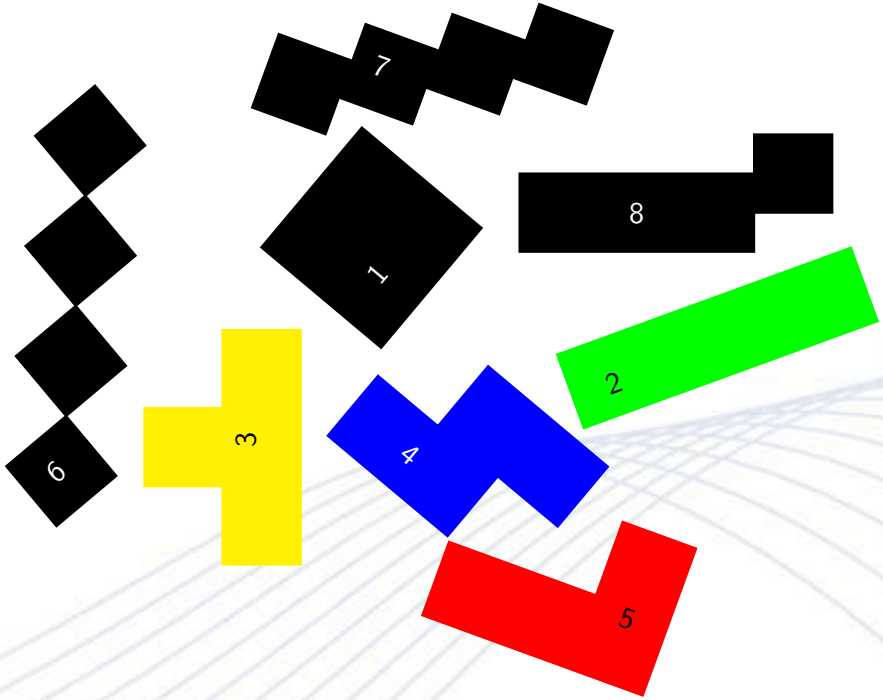
Pattern Recognition - Simple recognition task

Object	Area	Perimeter	Corners
1	4	8	4
2	4	10	4
3	4	10	8
4	4	10	8
5	4	10	6
6	4	16	16
7	4	13	16
8	4	11	8



Pattern Recognition - Simple recognition task

Object	Area	Perimeter	Color
1	4	8	black
2	4	10	green
3	4	10	yellow
4	4	10	blue
5	4	10	red
6	4	16	black
7	4	13	black
8	4	11	black



Business models with Big Data

Business models - General

- The basic logic of a company that describes how benefits for customers and partners are made possible and how these benefits flow back as sales
- Consists of elements and their combination:
  - Customer dimension: customer segments, customer channels, customer relationship
  - Benefit dimension: services and benefits
  - Value creation dimension: Resources, skills and processes
  - Partner dimension: Partners, partner channels and partner relationships
  - Financial dimension: turnover and costs
- Benefits enable differentiation from competitors through competitive advantages
- Achieving competitive advantages by combining and mutually strengthening the elements

Business models - General key questions I

Key partner

- Do I really need all partners?
- How could I save money/time?
- Are there comparable cheaper partners?
- What do I expect from my partners and what do they actually achieve?

Key activities

- What is my main business?
- Which of my activities contribute to success?

Key resources

- What resources are fundamental to my business and what is my current price-performance ratio?

Business models - General key questions II

Value offer

- What does my product or service promise and what does it achieve in the end?

Customer relationship

- Individual advice?
- Complete automation?
- Binding customers or just generating them?

Channels

- How do I service my customers?
- How do I best reach them?

Business models - General key questions III

Customers

- Do I serve "niche customers" or the mass market?

Cost causer

- Where exactly do costs actually arise?
- Are certain costs absolutely necessary?
- How could I reduce costs?

Sources of income

- How is the turnover generated?
- Are there other ways to generate revenue?

Business models - General key questions III

Constantly questioning and clearly defining elements of the business model in order to expand competitive advantages

Business models with Big Data

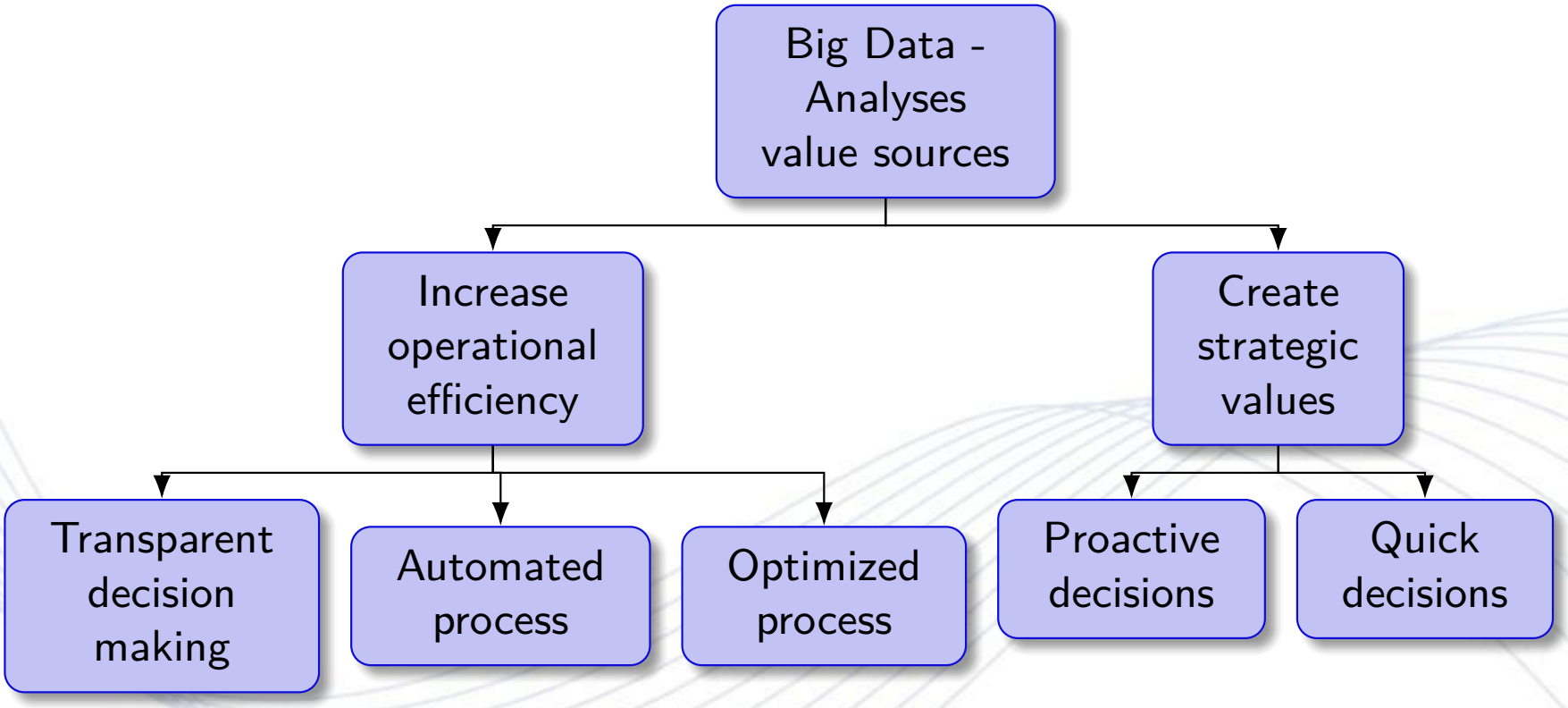
Data-based business models

- Business model innovation: New combination of elements to create value for customers and partners, build competitive advantage and generate growth
- Owning large amounts of data does not yet create added value, only value-adding applications generate benefits

Value creation builds on a business model

Business models with Big Data

Added value of Big Data analyses



Business models with Big Data

Development of a data-based business model:

1. Optimization of an existing model
2. Extension of an existing model
3. New conception of a model

Business models with Big Data

Optimization of an existing business model by Big Data

Example: Lift and escalator manufacturer

- Problem: Failure of elevators is only noticed and reported to the company late
- Solution: at standstill, errors detected by sensors are reported to the control center, service technicians are informed via app and can have suitable spare parts ready
- Advantage of Big Data: Predictive Maintenance enables early detection, avoidance or elimination of faults
- Customer benefit: reliable availability of the elevators

Business models with Big Data

Expansion of an existing business model through Big Data

- Problem: high investments for infrequent use of elevators
- Solution: use-dependent tariff for elevators, few users could pay "per trip" without having to lower the regular market price
- Advantage of Big Data: additional services tailored to the user profile of frequent users can be offered, dynamic price models are possible
- Customer benefits: usage-based pricing could acquire new customers, as high investments would be eliminated; tailored customer and maintenance service

Business models with Big Data

New conception of a completely data-based business model

- Problem: efficient design of an agricultural enterprise
- Solution: Platform for monitoring agriculture to support decision-making and increase efficiency and resource utilization
- Advantage of Big Data: heterogeneous data such as machine data from different manufacturers, data on soil properties or weather data are stored and evaluated
- Customer benefits: Vendor-independent evaluation of data, holistic overview for the customer, increased transparency and avoidance of silos

Business models with Big Data

Identify potential of Big Data

- Change in own business model leads to added value of data
- Regardless of the attractiveness of the existing model, digitization and further developments are changing the added value of products and services
- Disruptions displace old technologies and offers, e.g.
  - Kodak with analog film rolls, replacement by digital cameras
  - Encyclopedias like Brockhaus, replacement by online databases
  - Classified ads in newspapers, replacement by online platforms
- Improvement through digitization and Big Data, e.g.
  - cab centers, Moia, Uber or Ola
  - Mail order business

Business models with Big Data

Fully integrate Big Data capabilities to generate maximum value